

Example 4: In the production of tyres of an automobile, the output of a given size was inspected every day prior to the tyres being given to finished goods stores. The number of defective tyres found every day inspection is summarised in the following table:

Number of Sample	Number of Tyres Inspected	Number of Defective Tyres	Number of Sample	Number of Tyres Inspected	Number of Defective Tyres
1	650	70	11	670	71
2	510	74	12	660	75
3	600	58	13	600	77
4	590	61	14	550	78
5	630	65	15	540	64
6	650	115	16	610	90
7	700	82	17	670	96
8	740	55	18	660	110
9	580	80	19	650	78
10	600	90	20	590	60

Draw the appropriate control chart and comment on the state of the process.

Number of Sample	Number of Tyres Inspected (n)	Number of Defective Tyres (d)	Proportion of Defective Tyres (p = d/n)
1	650	70	0.108
2	510	74	0.145
3	600	58	0.097
4	590	61	0.103
5	630	65	0.103
6	650	115	0.177
7	700	82	0.117
8	740	55	0.074
9	580	80	0.138
10	600	90	0.150
11	670	71	0.106
12	660	75	0.114
13	600	77	0.128
14	550	78	0.142
15	540	64	0.119
16	610	90	0.148
17	670	96	0.143
18	660	110	0.167
19	650	78	0.120
20	590	60	0.102
Total	12450	1549	

From the above table, we have

$$\bar{p} = \frac{\sum_{i=1}^k d_i}{\sum_{i=1}^k n_i} = \frac{1549}{12450} = 0.124$$

According to this approach, the control limits are calculated using the average sample size and we get constant control limits just as we get in the case of constant sample size. We can calculate the average sample size using equation (14) as follows:

$$\bar{n} = \frac{1}{k} \sum_{i=1}^k n_i = \frac{1}{20} \times 12450 = 622.5$$

Using equations (16a to 16c), we calculate the centre line and control limits as follows:

$$CL = \bar{p} = 0.124$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.124 + 3\sqrt{\frac{0.124 \times (1-0.124)}{622.5}}$$

$$= 0.124 + 3 \times 0.013 = 0.163$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.124 - 3\sqrt{\frac{0.124 \times (1-0.124)}{622.5}}$$

$$= 0.124 - 3 \times 0.013 = 0.085$$

Then, we plot the p-chart (see Fig. 3.4).

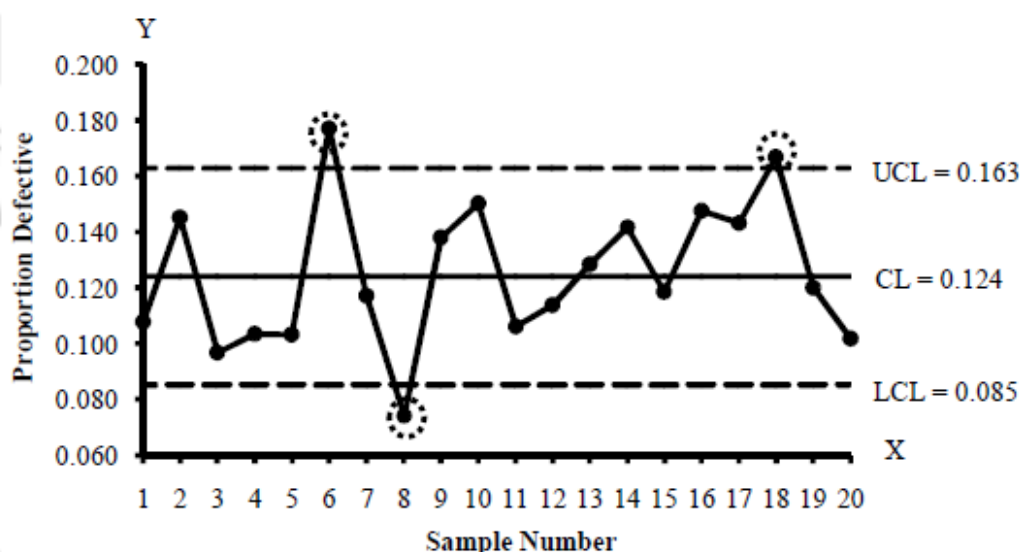


Fig. 3.4: The p-chart for proportion of defective tyres.

Interpretation of the result

From Fig. 3.4, we note that the points corresponding to sample numbers 6, 8 and 18 lie outside the control limits. Therefore, the process is out-of-control.

Hence, both approaches give the same conclusion about the process.

Now, you can try the following exercise and draw the p-chart for variable sample size.